$$c_0 + c_1(x-a) + c_2(x-a)^2 + c_3(x-a)^3 + \cdots$$

$$\parallel$$

Power Series

Standard form: $\sum c_n (x-a)^n$

Other forms: $\sum c_n (ax+b)^n$.

$\sum c_n (x-a)^n$ is a power series centered at a.

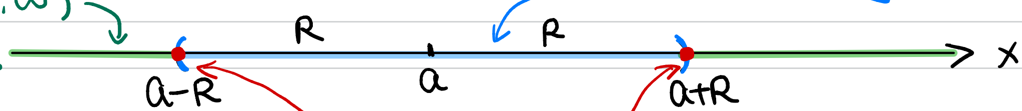
Radius of convergence is a number R s/t the series $\sum c_n (x-a)^n$ diverges when $|x-a| > R$, and the series converges absolutely when $|x-a| < R$. The series may or may not converge at the endpoints, i.e. $|x-a| = R$.

Interval of convergence is the interval for values of x s/t the series converges and the series diverges outside this interval.

e.g. for $\sum c_n (x-a)^n$ w/ radius of convergence R.

$(-\infty, a-R) \cup (a+R, \infty)$ the series diverges

the series converges absolutely on $(a-R, a+R)$ $\Rightarrow$ converges



At the two endpoints, the series may or may not converges (conditionally) (You will need to check this separately)

Interval of Convergence: "[" or "(" $a-R$, $a+R$ ")" or "]"

Taylor Polynomials

DEFINITIONS Let f be a function with derivatives of all orders throughout some interval containing a as an interior point. Then the **Taylor series generated by f at $x = a$** is

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \cdots .$$

DEFINITION Let f be a function with derivatives of order k for $k = 1, 2, \ldots, N$ in some interval containing a as an interior point. Then for any integer n from 0 through N, the **Taylor polynomial of order n** generated by f at $x = a$ is the polynomial

$$P_n(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(k)}(a)}{k!}(x-a)^k + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n.$$

Taylor Polynomials

Taylor's Formula

If f has derivatives of all orders in an open interval I containing a, then for each positive integer n and for each x in I,

$$f(x) = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots$$

$$+ \frac{f^{(n)}(a)}{n!}(x-a)^n + R_n(x), \qquad (1)$$

where

$$R_n(x) = \frac{f^{(n+1)}(c)}{(n+1)!}(x-a)^{n+1} \quad \text{for some } c \text{ between } a \text{ and } x. \qquad (2)$$

$$\Rightarrow |R_n(x)| \leq \max_{c\in[a,x]}\left(|f^{(n+1)}(c)|\right)\frac{(x-a)^{n+1}}{(n+1)!}$$

Taylor Series : MacLaurin Series

DEFINITIONS Let f be a function with derivatives of all orders throughout some interval containing a as an interior point. Then the **Taylor series generated by f at $x = a$** is

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(a)}{k!}(x-a)^k = f(a) + f'(a)(x-a) + \frac{f''(a)}{2!}(x-a)^2 + \cdots + \frac{f^{(n)}(a)}{n!}(x-a)^n + \cdots.$$

The **Maclaurin series of f** is the Taylor series generated by f at $x = 0$, or

$$\sum_{k=0}^{\infty} \frac{f^{(k)}(0)}{k!}x^k = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \cdots + \frac{f^{(n)}(0)}{n!}x^n + \cdots.$$

e.g. MacLaurin Series for sin(x):

$f'(x) = \cos(x) \qquad f''(x) = -\sin(x) \qquad f'''(x) = -\cos(x) \qquad f^{(4)}(x) = \sin(x) \quad \cdots$

$f'(0) = 1 \qquad f''(0) = 0 \qquad f'''(0) = -1 \qquad f^{(4)}(x) = 0 \quad \cdots$

Hence $\sin(x) = \cancel{f(0)} + f'(0)x + \cancel{\frac{f''(0)}{2!}x^2} + \frac{f'''(0)}{3!}x^3 + \cancel{\frac{f^{(4)}(0)}{4!}x^4} + \cdots$

$$= \frac{1}{1!}x + \frac{-1}{3!}x^3 + \frac{1}{5!}x^5 + \frac{-1}{7!}x^7 + \cdots$$

$$= \sum_{k=0}^{\infty} (-1)^k \frac{x^{2k+1}}{(2k+1)!}$$

e.g. MacLaurin Series for e^x:

$f'(x) = e^x \qquad f''(x) = e^x \qquad f'''(x) = e^x \qquad f^{(4)}(x) = e^x \quad \cdots$

$f'(0) = 1 \qquad f''(0) = 1 \qquad f'''(0) = 1 \qquad f^{(4)}(0) = 1 \quad \cdots$

Hence $e^x = f(0) + f'(0)x + \frac{f''(0)}{2!}x^2 + \frac{f'''(0)}{3!}x^3 + \frac{f^{(4)}(0)}{4!}x^4 + \cdots$

$$= 1 + 1\cdot x + \frac{1}{2!}x^2 + \frac{1}{3!}x^3 + \frac{1}{4!}x^4 + \cdots$$

$$= \sum_{k=0}^{\infty} \frac{x^k}{k!}$$